AI CAREERS FOR WOMEN

LINKEDIN FELLOWSHIP

Online Store Inventory and Order Fulfilment Analysis

Graphic Era Deemed to be University

Project Domain

[Excel with Copilot]

A Collaborative Initiative

equipping students with practical skills in AI, Data Analytics, and Business Intelligence

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Year - 2026

3. Problem Statement

Online stores handle large volumes of inventory and order data daily. However, this data often contains issues such as duplicate records, missing values, inconsistent formats, and inaccurate stock information.

Without proper data cleaning and analysis, businesses cannot effectively track inventory, manage orders, or make informed decisions. This leads to problems like stock shortages, delayed deliveries, and poor customer satisfaction.

This project addresses these challenges by applying data cleaning, analysis, and visualisation techniques to transform raw data into meaningful insights.

4. Objectives

- To collect and understand the online store inventory dataset
- To clean and preprocess data by removing duplicates and handling missing values
- To analyse inventory levels, order trends, and delivery performance
- To create dashboards and visualisations using Excel
- To integrate AI tools like Copilot/ChatGPT for insight generation
- To present findings in a clear and structured format

5. Methodology

#	Phase	Description
1	**Data Collection & Exploration**	Gather the dataset and understand the structure, fields, and data quality.
2	**Data Cleaning & Preprocessing**	Remove duplicates, handle missing values, fix formats
3	**Analysis & AI Integration**	Use Excel formulas, pivot tables, and AI tools for insights.
4	**Visualisation / Output**	Create dashboards, charts, and reports

5	**Documentation & Submission**	Prepare report, PPT, and project submission

6. Resources and Timeline

A. Hardware & Software Requirements

- Laptop or PC — Intel Core i3
- Operating System: Windows 10
- Microsoft Excel (Excel 2019)
- Microsoft Copilot — via Microsoft 365 subscription
- AI Chatbot (Microsoft Copilot Web)

Internet connection — required for Copilot and AI Chatbot access

- Web browser — Google Chrome or Microsoft Edge (latest version)

B. Dataset

- The dataset was not directly available in a ready-to-use format and was partially sourced and manually generated.
- It is inspired by datasets from platforms like Kaggle (https://www.kaggle.com) and Excel sample data.
- The dataset includes around 660 rows and 14 columns
- It contains fields like Product ID, Category, Stock Level, Orders, Delivery Status, etc.
- The dataset includes intentional issues such as duplicates, missing values, and formatting errors
- Data cleaning and analysis were performed using Excel and AI tools. Project Timeline

Phase	Task	Tentative Completion Date
1	Dataset collection, exploration, and understanding	*Week 1*
2	Data cleaning, preprocessing, and validation	*Week 2*
3	Analysis, AI tool integration (Copilot), and insight generation	*Week 2 – 3*
4	Dashboard/output build, testing, and review	*Week 3*
5	Report writing, GitHub upload, and video recording and submission	*Week 4*

7. Conclusion

This project demonstrates how Excel and AI tools can be used to analyse inventory and order data effectively. After cleaning the dataset, key insights were identified related to product demand, stock levels, and delivery performance.

The project highlights the importance of data cleaning and visualisation in improving business decision-making. It shows that combining data analytics with AI tools can significantly enhance efficiency and provide valuable insights for e-commerce operations.

8. References

[1] Microsoft Corporation. Microsoft Excel Documentation. Available at: https://support.microsoft.com/en-us/excel

[2] Microsoft Corporation. Microsoft Copilot — AI-Powered Productivity. Available at: https://copilot.microsoft.com

[3] Edunet Foundation. AICW Fellowship Program Guidelines. 2026.